

1. Administrative & Study Identification

Full Study Title: An Open-label, Multi-center, Phase 1/2 Study to Assess Safety, Cellular Kinetics and Exploratory Efficacy of Rapcabtagene Autoleucel in Participants With Difficult-to-treat Rheumatoid Arthritis and Severe, Refractory Sjogren's Disease With Organ Involvement **Short Title/Acronym:** A Study to Assess Safety, Cellular Kinetics and Exploratory Efficacy of Rapcabtagene Autoleucel in Rheumatoid Arthritis and Sjogren's Disease **Protocol Number:** CYTB323M12101B **NCT Number:** NCT07048197 **Posting ID:** 465314049293962271 **Trial Status:** RECRUITING **Sponsor / Company:** Novartis Pharmaceuticals / Novartis AG **Investigational Product:** Rapcabtagene Autoleucel (YTB323 / Anti-CD19 CAR-T Cell Therapy) **Phase of Development:** Phase 1, Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, cellular kinetics, and any early sign of efficacy of rapcabtagene autoleucel in participants with difficult-to-treat rheumatoid arthritis (D2T RA) and severe, refractory Sjogren's disease (srSjD) with organ involvement. **Secondary Objectives:** To evaluate long-term tolerability, specific B-cell depletion/recovery dynamics, and exploratory clinical efficacy/disease remission metrics following treatment.

2.2 Background & Rationale To address the significant unmet medical need in patients with severe active rheumatoid arthritis and Sjogren's disease who have experienced treatment failure with standard of care therapies. Research suggests that CD19-directed CAR-T cell therapies like rapcabtagene autoleucel can quickly and profoundly deplete B cells, potentially resetting the immune system and helping the autoimmune disease go into long-term remission.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel Assignment (Evaluating multiple disease cohorts without a placebo control) **Blinding:** Open-label (None) **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 27 participants **Number of Sites:** Multi-center (includes sites in

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent must be obtained prior to participation in the study. 2. Men and women aged ≥ 18 years and ≤ 75 years at Screening. 3. **For RA only:** Diagnosis of rheumatoid arthritis; Treatment failure of standard of care therapies; Active disease. 4. **For SjD only:** Diagnosis of Sjogren's disease; Active disease.

4.2 Exclusion Criteria 1. BMI at Screening of ≤ 17 or ≥ 40 kg/m². 2. Clinically significant active, opportunistic, chronic or recurrent infection, or inadequate organ function during screening. 3. History of lymphoproliferative disease, any known malignancy, or history of bone marrow/hematopoietic stem cell or solid organ transplantation. 4. Sexually active males unwilling to use a condom during intercourse from the time of enrollment. 5. Women of childbearing potential, unless they are using a highly effective method of contraception starting from the time of enrollment. 6. Female participants who are pregnant, breastfeeding or intending to conceive during the course of the study. 7. Any psychiatric condition or disability making compliance with treatment or informed consent impossible.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose/infusion of rapcabtagene autoleucel (CD19-directed CAR-T cells) post-lymphodepletion. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Single administration followed by a long-term observational follow-up period up through 2028. (Note: Related CAR-T structured data mapping for reference identifies the drug name as ciltacabtagene autoleucel, Trade Name: Carvykti, ATC Code: L01XL05, EMA URL: https://www.ema.europa.eu/en/documents/product-information/carvykti-epar-product-information_en.pdf).

5.2 Concomitant & Prohibited Therapy Permitted: Standard bridging or supportive medications as dictated by the protocol to manage the underlying autoimmune conditions prior to conditioning. **Prohibited:** Concurrent investigational therapies, live virus vaccinations, or systemic therapies that may excessively suppress the expansion or viability of the infused CAR-T cells.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent,

Medical History, Organ Function Assessment, Infection Screening |
| **Baseline** | Day 0 | Leukapheresis, Lymphodepleting Chemotherapy,
CAR-T Infusion | | **Interim** | Weeks 1-12 | Acute Safety Labs (CRS/
ICANS monitoring), Cellular Kinetics (qPCR), Efficacy Assessment
| | **End of Study** | Up to Nov 2028 | Final Long-term Vital Signs,
Exit Interview, Disease Remission Assessment |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Number of Participants with Adverse Events as a Measure of Safety and Tolerability: Safety parameters include vital signs, adverse events, laboratory parameters, and ECG evaluation.

7.2 Secondary & Exploratory Endpoints 1. **Pharmacokinetic (PK) parameters:** Maximum observed blood concentration (C_{max}), Time to reach maximum concentration (T_{max}), and Terminal elimination half-life ($T_{1/2}$). 2. Clinical efficacy and disease activity scoring for Rheumatoid Arthritis and Sjogren's Disease. 3. Characterization of peripheral B-cell depletion and subsequent repopulation timelines.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring for CAR-T specific toxicities, including Cytokine Release Syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS), cytopenias, and infections. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent oversight committee utilized due to the Phase 1/2 nature and advanced cellular therapy risks.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received the rapcabtagene autoleucel infusion); Cellular Kinetics Set. **Sample Size Justification:** Target enrollment of $N=27$ is based on the exploratory Phase 1/2 nature of the trial, sufficiently powered to evaluate initial safety profiles, dose tolerability, and cellular kinetic parameters. **Handling of Missing Data:** Standard handling for PK profiles; non-responder imputation (NRI) or Last Observation Carried Forward (LOCF) for early withdrawals regarding efficacy markers.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Rigorous onsite and remote monitoring tailored for advanced cellular therapies (ATMP), ensuring rapid reporting of severe safety events. **Audit Trail:** Assured via validated electronic case report forms (eCRF) and centralized laboratory tracking for cellular kinetics.

11. Study Identifiers & Governance

Primary ID: CYTB323M12101B **Registry Number:** NCT07048197 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Study to Assess Safety, Cellular Kinetics and Exploratory Efficacy of Rapcabtagene Autoleucel in Rheumatoid Arthritis and Sjogren's Disease **Official Regulatory Title:** An Open-label, Multi-center, Phase 1/2 Study to Assess Safety, Cellular Kinetics and Exploratory Efficacy of Rapcabtagene Autoleucel in Participants With Difficult-to-treat Rheumatoid Arthritis and Severe, Refractory Sjogren's Disease With Organ Involvement

12. Clinical Framework & Disease Classification

Disease Areas: Rheumatoid Arthritis (RA) and Sjögren's Disease (SjD) **Condition Name / Preferred Name:** Arthritis **Preferred UMLS Name:** Autoimmune Diseases **Semantic Type:** Disease or Syndrome **MeSH Code:** D001168 **HPO Code:** HP:0001369 **SNOMED ID:** 3723001 **Medical Methodology:** Autologous CAR T-cell Therapy (B-cell depletion) **Optimization Target:** Rapid and sustained reduction of CD19-positive B cells to reset immune activity

13. Study Procedures & Methodology

Management Pathway: Standard CAR-T clinical pathway (Apheresis \$ \rightarrow \$ Manufacturing \$ \rightarrow \$ Lymphodepletion \$ \rightarrow \$ Infusion \$ \rightarrow \$ Monitoring). **Education Protocol (HCP):** Mandatory site training on handling rapcabtagene autoleucel, and the identification/management of CRS and neurotoxicity. **Education Protocol (Patient):** Onsite training regarding CAR-T expectations, reporting fevers/neurologic symptoms immediately, and long-term infection prevention. **Quality Audit Mechanism:** Cellular kinetics tracking (qPCR) & central safety auditing for dose-limiting toxicities.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up extending to study completion in November 2028 (approx. 3.5 years) **Onsite Visit Cadence:** Intensive monitoring in the first 28 days post-infusion, followed by monthly, then quarterly evaluations. **Remote Monitoring Frequency:** Routine remote check-ins for late-onset adverse events or

infection tracking. **Checklist Review Interval:** Regular interval reviews for disease activity indices specific to RA and SjD.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				